MILESTONE 1 REPORT

Formula 1 Podium Prediction

Predicting top-3 finishes from information available after qualifying and before the race start

Team	[team members and IDs]
Data	Formula 1 World Championship (1950–2024), Kaggle, 14 tables
Notebook	[Kaggle link]
Code	[GitHub link]

Key results

- A leakage-safe modeling table with one row per driver per race (26,668 rows; 24,586 after removing non-starters and the Indianapolis 500) and 17 pre-race features in four groups: grid, form, standings and qualifying.
- Time-based split: train 1950–2019, validation 2020–2021, test 2022–2024. All choices were made on training and validation data only.
- On the unseen test seasons, every model beats the grid-only baseline (PR-AUC 0.647): logistic regression 0.720, gradient boosting 0.701 (the model selected on validation), neural network 0.692.
- Grid position dominates every analysis (single-feature AUC 0.890, ablation, permutation importance, SHAP). Form, standings and qualifying add value together but largely overlap.
- Data-engineering answers: front-row conversion rankings by circuit are not stable across eras (Spearman −0.03); there is no home advantage once grid position is controlled for (actual/expected 0.98); mechanical retirements fell from 9.6% to 5.6% of starts, with Alfa Romeo and Alpine highest in 2022–2024.

1. Task and prediction cutoff
2. Data audit and cleaning
3. Exploratory data analysis
4. Data-engineering questions
5. Modeling table and features
6. Train, validation and test split
7. Preprocessing
8. Models and evaluation
9. Experiments
10. Explainability
11. Inference function
12. Known limitations

1. Task and prediction cutoff

For every driver in a race, we predict whether he finishes in the top three (`positionOrder` ≤ 3). This is a binary classification task. The prediction is made **after qualifying and before the race start**. Every feature must therefore be known at that moment: the starting grid, qualifying results, and results from *earlier* races. Anything produced by the race itself (finishing position, points, laps, status, race time, fastest lap, pit stops, lap times, and the standings after the race) is either the target or forbidden.

The rule we followed throughout: post-race information may be used only as the target, or as history from earlier races. An assertion in the notebook checks that no post-race column is in the feature list.

2. Data audit and cleaning

2.1 Audit of the raw tables

The 14 CSV files are loaded once into a dictionary and never modified; all cleaning works on copies. The audit checked four things.

- **Sentinels.** The dataset stores missing values as the text `\N`. Pandas therefore reported 0 missing cells in `results`, while 122,887 cells held `\N`. The placeholders occur in 30 columns (Figure 1). Most mean "does not apply": `results.position` (40.9%) belongs to unclassified drivers, `qualifying.q2` / `q3` (44.1%/65.0%) to drivers eliminated earlier or to one-session formats, and the race-time and fastest-lap columns (68–71%) are post-race information.
- **Primary keys** are unique and non-null in all 14 tables.
- **Foreign keys:** 23 relationships checked, 0 orphan rows, so no join loses rows.
- **Grain:** 176 result rows cover 85 (raceId, driverId) pairs in which a driver appears more than once (Figure 2).

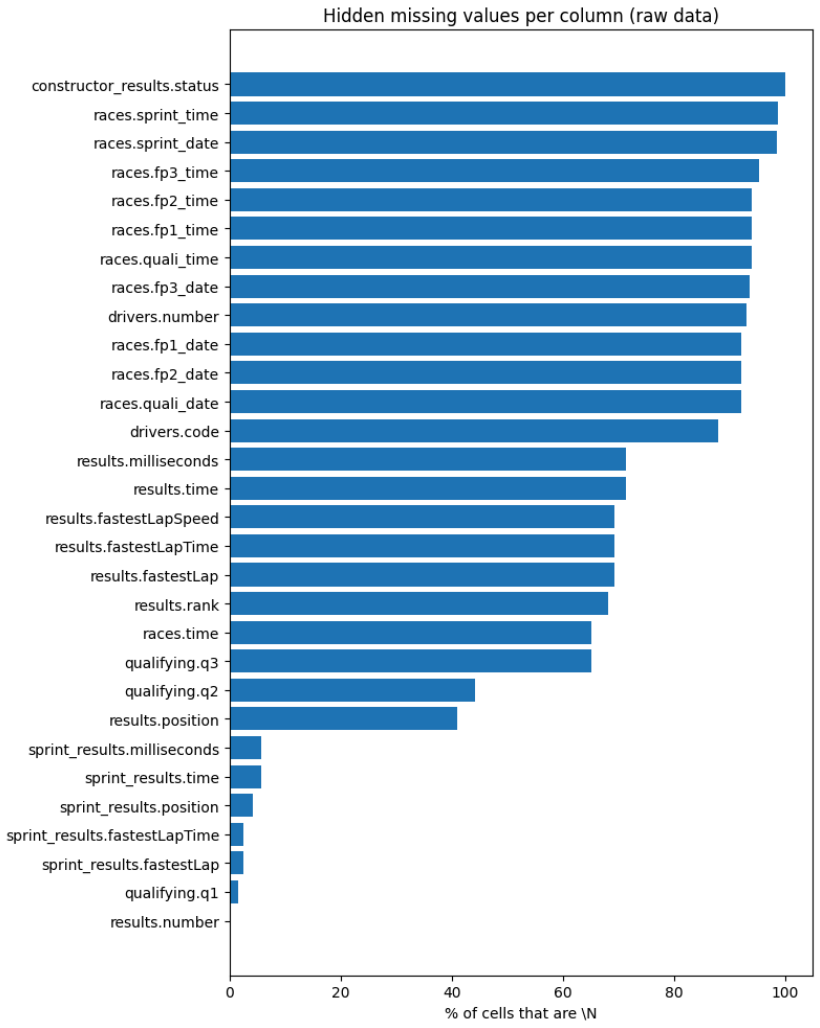


Figure 1. Share of `\N` placeholders per column in the raw data.

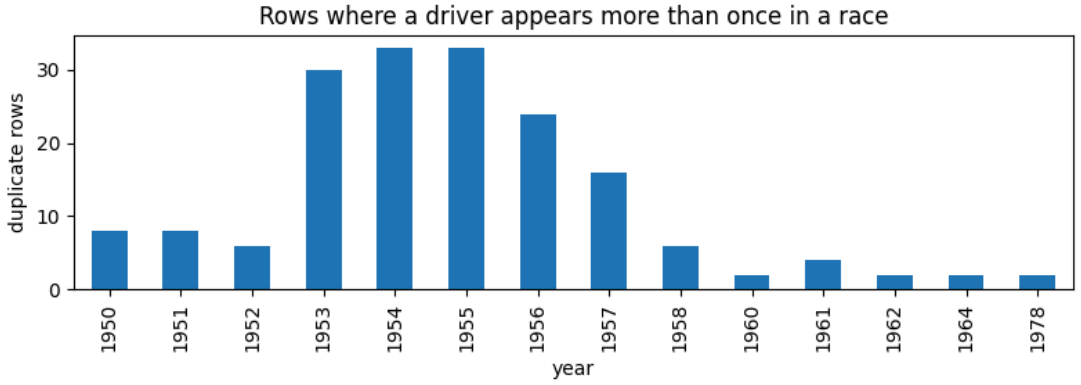


Figure 2. Result rows in which a driver appears more than once in a race, by season.

2.2 Sentinel handling and semantic typing

`\N` was replaced by real missing values (0 remain in the cleaned copy; all 122,887 remain in the raw table). Figure 3 shows the effect: before cleaning pandas saw no missing values in `results`; afterwards the true gaps appear. Columns stored as text were converted to their real types: numbers, dates, and qualifying lap times such as `1:26.572` to seconds (86.572). All q1–q3 times converted (0 unreadable values).

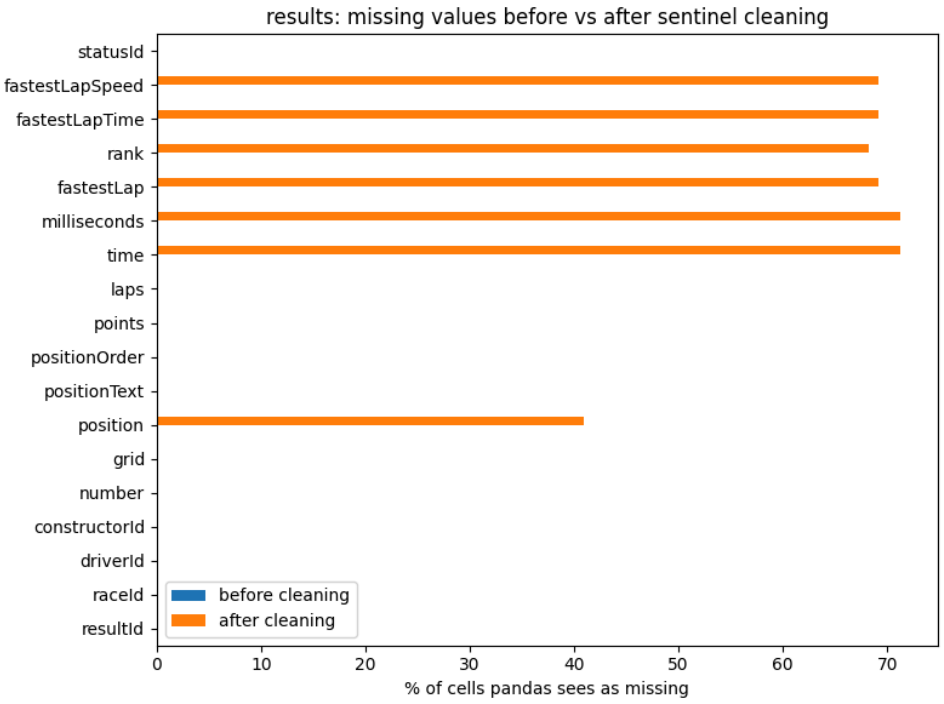


Figure 3. Share of cells pandas recognises as missing in `results`, before and after sentinel cleaning. The "before" bars are all at 0%.

2.3 One row per driver per race

Most duplicates are shared drives from 1950–1957, when a driver could take over a teammate's car during the race. Nine pairs come later (1958–1978): three are drivers entered in two cars who never started, and six are car changes (Figure 2).

Rule: keep the row with the best (lowest) `positionOrder`. This removed 91 rows and left 26,668, with (raceId, driverId) verified unique.

- The target asks whether the driver finished on the podium; if any of his cars was classified in the top three, he was.
- `positionOrder` is never missing, so the rule always gives an answer.
- The alternative "keep the car he started in (non-zero grid)" cannot be applied: both rows usually have a non-zero grid.

After de-duplication, 1,107 races have exactly three podium rows, 16 have four and 2 have five (two drivers who shared a podium car are both classified), giving 3,395 podium rows.

2.4 What grid = 0 means

`grid` has no `\N`, but uses 0 for "no grid slot": 1,635 rows (6.1%). Of these, 1,346 are F (failed to qualify) and 202 W (withdrawn); most of the remainder raced from the pit lane. No driver with grid 0 has ever finished on the podium. Grid 0 therefore cannot stay the number 0, which a model would read as better than pole; Section 5 explains the encoding.

3. Exploratory data analysis

Target and imbalance. 3,395 of 26,668 rows (12.7%) are podiums. Always predicting "no podium" would be 87% accurate, so accuracy is not used.

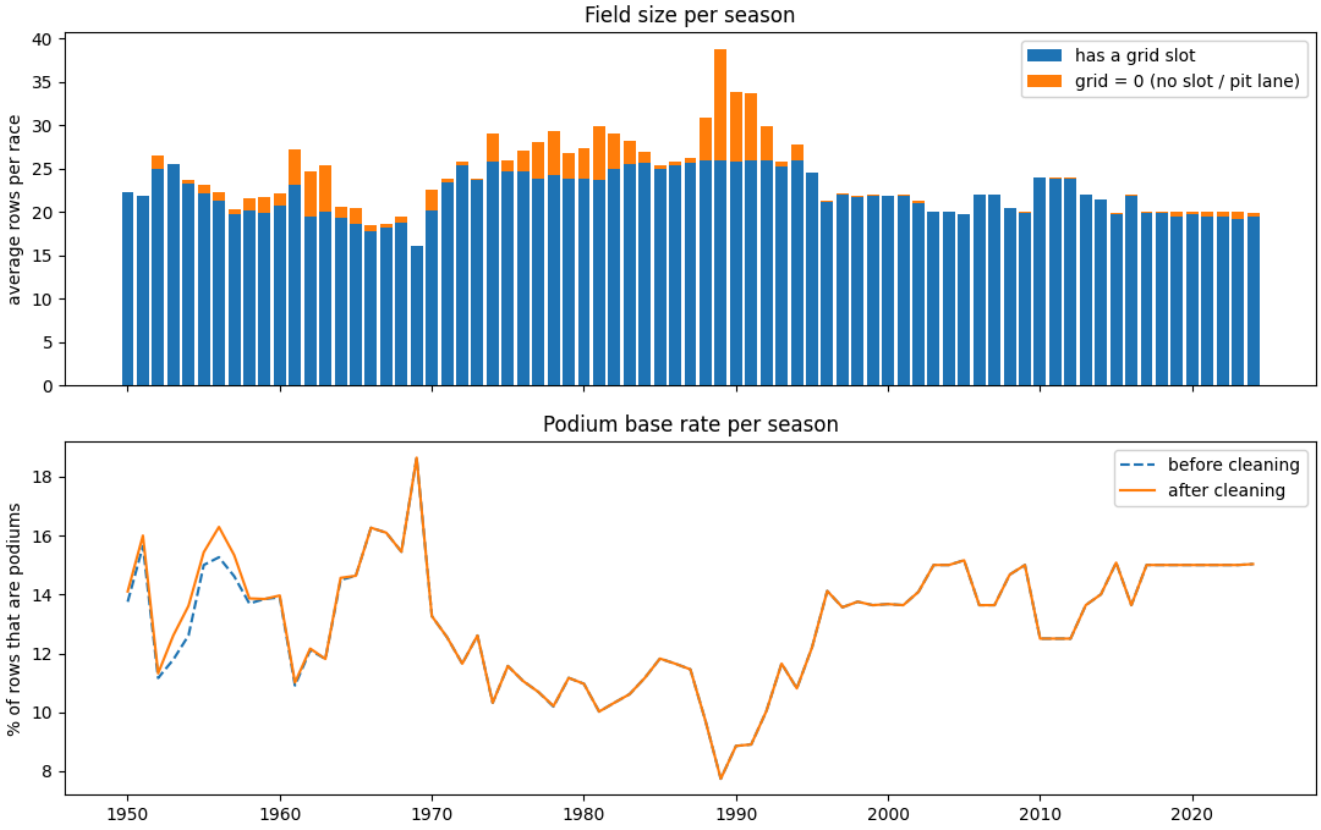


Figure 4. Top: average result rows per race, split into drivers with a grid slot and grid 0. Bottom: share of rows that are podiums per season, before and after cleaning.

- **Field size.** Starters per race are stable (about 16–26). Rows without a grid slot cluster in the early 1960s, the mid-1970s to early 1980s and above all 1988–1992 (1989: about 39 entries per race for 26 starters). These are drivers who never started; they are removed from the modeling population (Section 6).
- **Podium base rate.** Roughly three divided by the number of entries per race: 7.7% in 1989, 12.5% in 2010 (24 cars), 15.0% in 2019 and 2024 (20 cars). Removing duplicates raised the 1950s rates slightly and changed nothing afterwards.

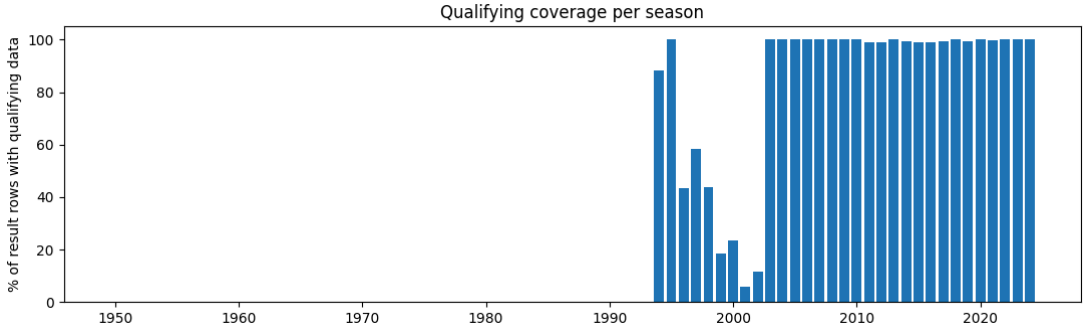


Figure 5. Share of result rows with a matching qualifying record, per season.

Qualifying coverage drifts strongly. There is no qualifying data before 1994, coverage is patchy in 1994–2002 and essentially complete from 2003; overall 39.4% of rows have it. Qualifying features are therefore missing for most old seasons.

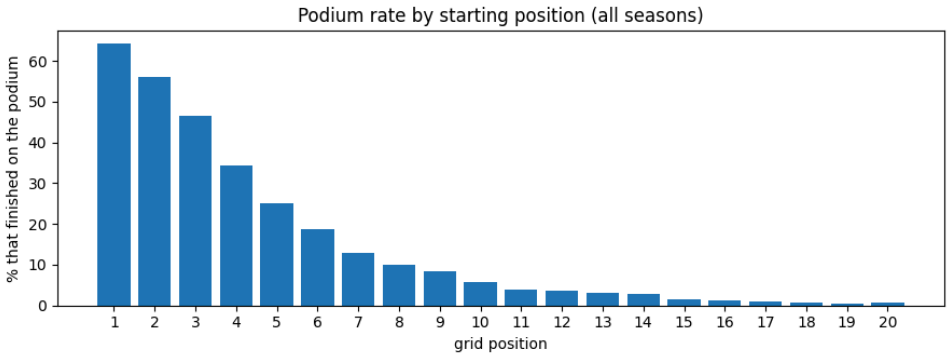


Figure 6. Podium rate by starting position, all seasons (about 1,120 rows per position).

Grid position. Pole leads to a podium 64.1% of the time, then 55.9%, 46.4%, 34.2%, 25.1% and 18.7% for positions 2–6, falling to almost zero by 15th. Grid is the strongest candidate feature. This is an association, not a cause: fast cars both qualify well and finish well.

4. Data-engineering questions

4.1 Which circuits convert a front-row start into a podium most often, and does the ranking change after 2014?

Method. Tables joined: results, races, circuits (and qualifying for the last part). Grain: one row per front-row starter per race (2,243 starts in 1,124 races). Front row = `grid` 1 or 2. A circuit is ranked only with at least 10 front-row starts in the era.

Rank	1950–2013	Rate	Starts	2014–2024	Rate	Starts
1	Magny-Cours	80.6%	36	Suzuka	88.9%	18
2	Istanbul Park	78.6%	14	Yas Marina	86.4%	22
3	Dijon-Prenois	75.0%	12	Monaco	85.0%	20
4	Bremgarten / Suzuka / Sepang	70.0%	10 / 50 / 30	Spa / Barcelona	81.8%	22
Last	(44 circuits ranked)			Hungaroring (19th of 19)	59.1%	22

Does the ranking change? Yes. Overall conversion rose from 56.0% to 76.0%, so we compare ranks rather than rates. Among the 16 circuits with enough starts in both eras, the Spearman rank correlation is **−0.03**: the two rankings are unrelated. Monaco rose from 15th to 3rd, Hungaroring fell from 2nd to 16th, and Bahrain stayed at 66.7% but dropped from 3rd to 14th. Only Suzuka is stable (1st in both).

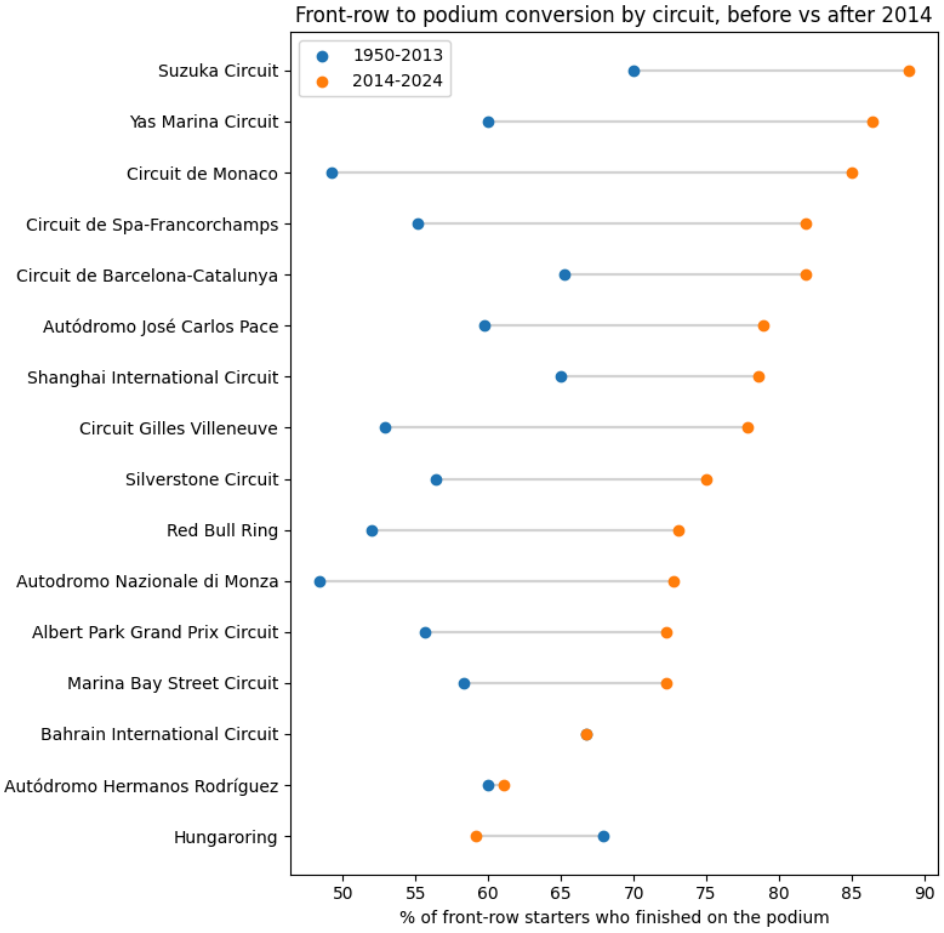


Figure 7. Front-row-to-podium conversion for the 16 circuits ranked in both eras (blue 1950–2013, orange 2014–2024).

Qualifying position vs grid. In 2003–2024, 38 of 856 front-row qualifiers (4.4%) did not start there, usually because of penalties. Their podium rate was 28.9%, against 72.2% for drivers who actually started on the front row (70.7% for all front-row qualifiers). The two columns differ, so we used `grid`, the position the car actually started from.

Missingness and cutoff. `grid` is complete for all seasons; qualifying data starts in 1994 and is complete only from 2003, so the qualifying comparison is limited to 2003–2024. Grid is known before the race; the analysis is descriptive. **Limitation:** hybrid-era circuits have only 14–26 front-row starts each, so one podium moves a rate by about 5 points; part of the reshuffling may be chance.

4.2 Do drivers podium more often at a circuit in their home country, after controlling for starting position?

Method. Tables joined: results, races, circuits (country), drivers (nationality). A nationality-to-country mapping defines a home race (2,490 rows, 9.3%). Rows with grid 0 are excluded because they have no starting position to control for. For each grid position we computed the podium rate of away drivers, and used it to calculate how many podiums home drivers would be expected to score from their grid positions.

	Home rows	Raw home rate	Raw away rate	Expected	Actual	Actual / expected
All races	2,322	11.5%	13.8%	271.2	266	0.98
Without Indianapolis 500 (main result)	1,942	11.8%	13.8%	234.7	230	0.98

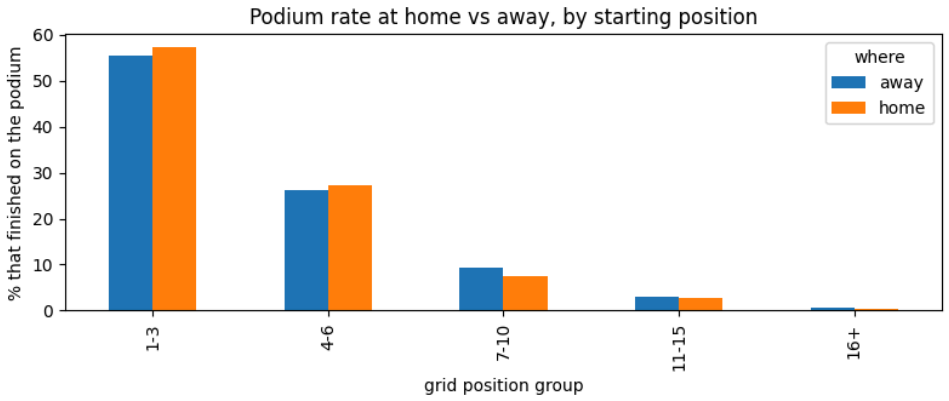


Figure 8. Podium rate at home and away within grid groups (without the Indianapolis 500). For grid 1–3: 57.2% home vs 55.5% away; 4–6: 27.1% vs 26.1%; 7–10: 7.4% vs 9.3%.

Answer: no home advantage once starting position is taken into account. The raw comparison makes home drivers look *worse* (11.5% vs 13.8%) because many home entries are local drivers at the back of the grid (737 home rows start 16th or lower). Controlled for grid, home drivers score 0.98 times the podiums expected, and within grid groups the rates differ by at most 2 points.

Indianapolis 500. These 11 races (1950–1960) have 381 rows, 99.7% of them home drivers, and supply most American home rows. A race without away drivers cannot inform a home/away comparison, so the version without it is the main result; the conclusion is the same either way. **Limitations:** nationality is not the same as where a driver lives; drivers from countries without a race (e.g. Finland) are always "away"; four rows with mixed nationalities are unmapped.

4.3 Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

Method. Tables joined: results, races, status, constructors. Grain: one row per race start in 2014–2024 (4,608 rows after removing 18 non-starters: status "Withdrew", or grid 0 with 0 laps). The 66 status values were grouped into *finished* (9), *mechanical* (40 component failures such as Engine, Power Unit, Gearbox, Brakes, Hydraulics), *other* (11: crashes, driver causes, disqualifications) and *borderline* (6: "Retired", Puncture, Tyre, Front wing, Rear wing, Undertray). Counting borderline cases as mechanical gives an upper rate. Constructors need at least 40 starts in the era.

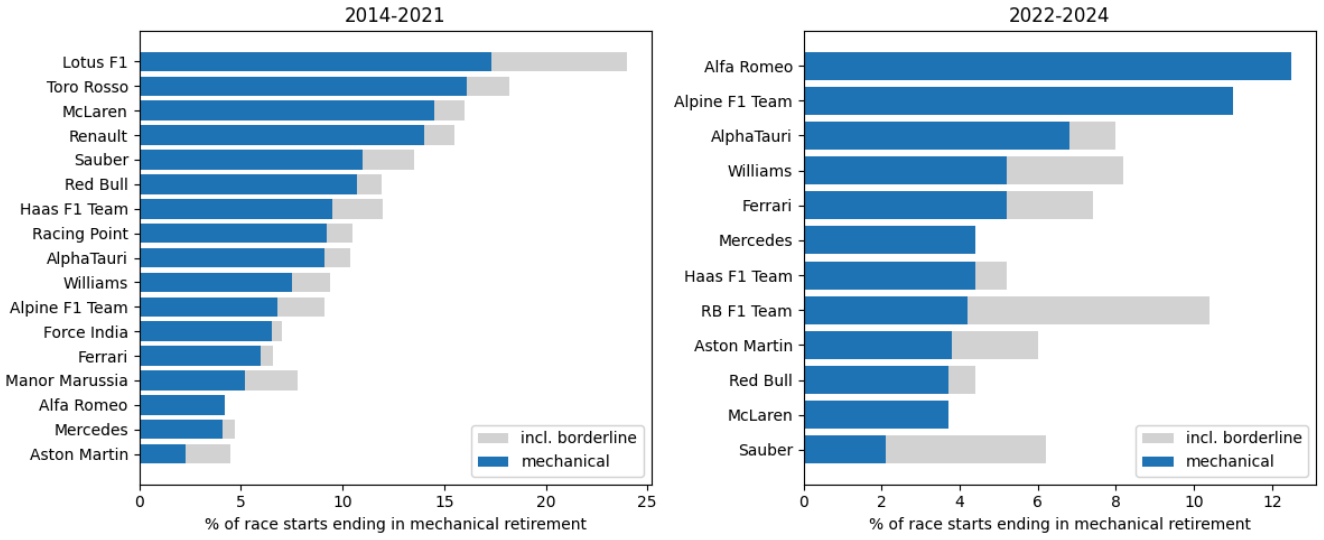


Figure 9. Share of race starts ending in a mechanical retirement per constructor. Coloured bar: mechanical; grey: including borderline statuses.

Era	Starts	Overall rate	Highest rates
2014–2021	3,255	9.6%	Lotus F1 17.3%, Toro Rosso 16.1%, McLaren 14.5%, Renault 14.0%
2022–2024	1,353	5.6%	Alfa Romeo 12.5%, Alpine 11.0% (about twice the era average)

Answer. Reliability improved overall. The leaders in both eras are unchanged when borderline statuses are counted as mechanical. McLaren improved from 14.5% to 3.7%, Alfa Romeo went from 4.2% to 12.5%, and Mercedes stayed low (4.1% and 4.4%). **Limitations:** renamed teams (Toro Rosso, AlphaTauri, RB) are separate entries because constructors follow `constructorId`; "Retired" gives no cause and is used unevenly (30 cases in 2014–2016, 9 in 2024), so the 2024-only entries RB (4.2%, up to 10.4%) and Sauber are uncertain; most 2022–2024 teams have fewer than 10 failures. `status` is post-race and is never a model feature.

5. Modeling table and features

Features were built on all 26,668 rows, so failed qualifying attempts remain part of a driver's history, and the population was chosen afterwards (Section 6). All history features use `shift(1)` or a join on the previous race, so a row never sees its own race.

Group	Feature	Meaning	Why it is known before the race
Grid	`grid_pos`	Starting position; grid 0 placed one slot behind the last car	Grid is fixed after qualifying and penalties
Grid	`grid_is_zero`	1 if no grid slot (pit-lane start)	Same
Grid	`field_size`	Number of cars on the grid	Same
Form	`drv_podium_last10`	Driver's podium rate in his previous 10 races	Earlier races only
Form	`drv_finish_last5`	Driver's average finishing order in his previous 5 races	Earlier races only
Form	`drv_podium_career`	Driver's podium rate over all earlier races	Earlier races only
Form	`drv_races_before`	Number of earlier races (0 on debut)	Count of earlier races
Form	`con_podium_last10`	Share of the team's car entries on the podium in its previous 10 races	Computed per team per race first, so a teammate's same-race result is excluded
Form	`con_races_before`	Number of earlier races of the team	Count of earlier races
Standings	`drv_std_pos`, `drv_std_points_share`	Championship position and points as a share of the leader's, before this race	Standings after the previous race of the same season
Standings	`con_std_pos`, `con_std_points_share`	The same for the team	Same
Qualifying	`quali_pos`	Qualifying position	Qualifying precedes the race
Qualifying	`quali_gap_pct`	Gap of the best qualifying lap to pole, in %	Same
Qualifying	`grid_drop`	Grid minus qualifying position (penalties)	Both known before the race
Qualifying	`has_quali`	1 if qualifying data exists	Data availability

Evidence for the choices (training seasons 1950–2019 only)

Feature (alone)	AUC	Missing	Feature (alone)	AUC	Missing
`grid_pos`	0.890	0.0%	`drv_std_pos`	0.845	14.5%
`con_podium_last10` (5 / 3 races)	0.846 (0.833 / 0.814)	1.4%	`con_std_points_share`	0.843	14.0%
`drv_podium_last10` (5 / 3 races)	0.838 (0.815 / 0.779)	3.5%	`drv_std_points_share`	0.839	14.5%
`drv_podium_career`	0.814	3.5%	win counts (dropped)	0.72–0.77	14%
`drv_finish_last5`	0.807	3.5%	`quali_pos` *	0.906	65.9%
race counts (driver / team)	0.668 / 0.631	0.0%	`quali_gap_pct` * / `grid_drop` *	0.887 / 0.544	66%

AUC of the feature used alone, in the stronger direction. *Measured only on the 34% of training rows with qualifying data (mostly recent seasons), so not directly comparable with grid.

- **Grid** is the strongest single feature; its value changed over time (Figure 10), from 53.0% podiums from pole in 1950–1993 to 80.7% in 2014–2024.
- **A 10-race window** beats 5 and 3 races for both driver and team, and beats the career rate: recent results matter more than old ones, but three races are too noisy.
- **Standings** use position and points share; raw points are not comparable across points systems, and win counts were weaker.
- **Qualifying position** correlates 0.975 with grid position; `quali_gap_pct` was capped (Section 7) because of broken laps up to 923%.

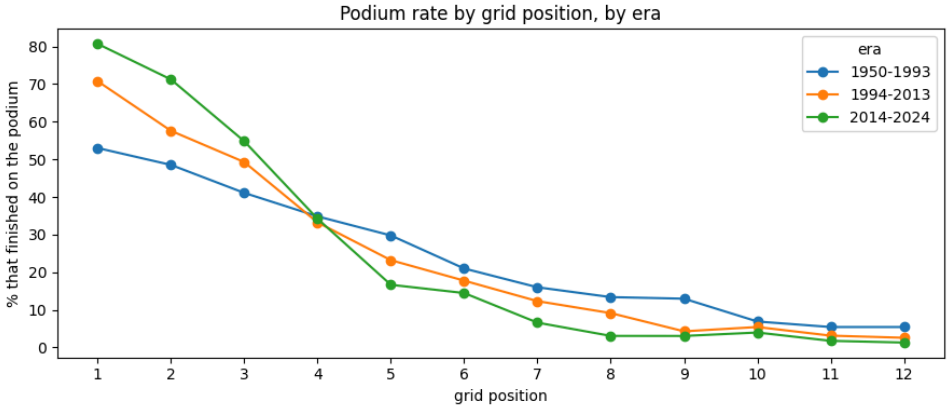


Figure 10. Podium rate by grid position in three eras. The curves cross at grid 4: in front of it the modern era converts more often, behind it less.

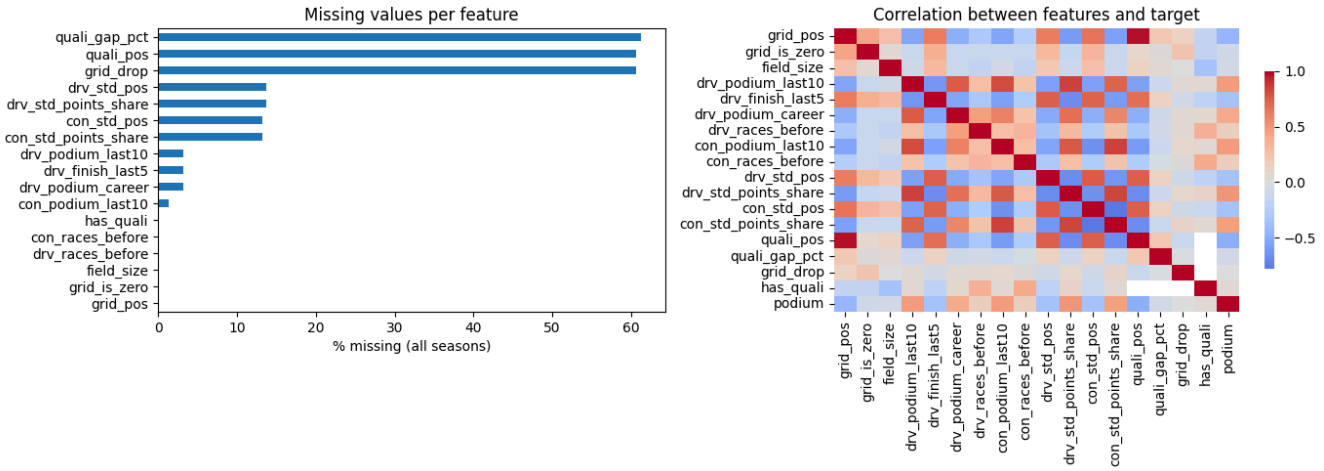


Figure 11. Left: missing values per feature (all seasons). Right: correlations between features and the target; form and standings form one strongly correlated block.

Leakage controls. A driver's debut has empty form; round 1 of every season has empty standings; before the final race of 2021 the two title contenders both have a points share of 1.0 (they were level on points), which is only possible with pre-race standings; an assertion confirms that no post-race column is a feature.

Not used. Current-race position, points, laps, status, race time, fastest lap, pit stops, lap times and post-race standings (all post-race). Sprint results are pre-race but exist only from 2021, so no training season contains them. Nationality and a home-race flag were left out because Question 2 found no home effect.

6. Train, validation and test split

Set	Seasons	Races	Rows	Podiums	Podium rate	Has qualifying
Train	1950–2019	1,007	22,459	3,038	13.5%	36.9%
Validation	2020–2021	39	775	117	15.1%	99.9%
Test	2022–2024	68	1,352	204	15.1%	100.0%

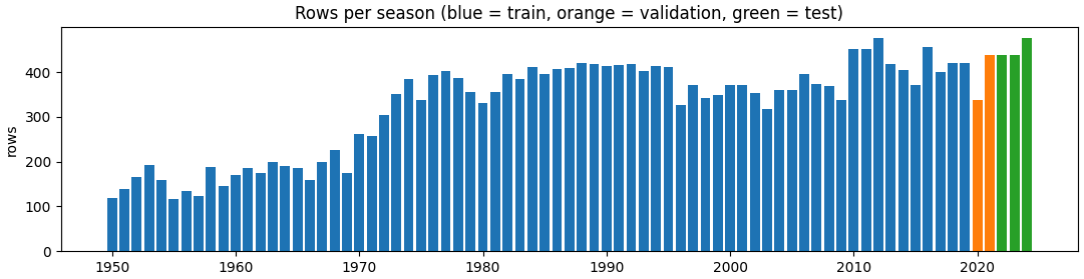


Figure 12. Modeling rows per season (blue train, orange validation, green test).

- **Split by season.** The task is to predict a race from earlier information. A random split would train on races after the ones being tested, and would separate drivers of the same race across sets although a race has only three podium places. Assertions confirm that no race appears in two sets and the sets are in time order.
- **Roles.** Training fits models; validation (the two seasons nearest the test period) selects hyperparameters, the training window and the probability threshold; the test set is used once, at the end.
- **Population (26,668 → 24,586 rows).** 1,701 non-starters (codes F and W, both known before the start) were removed: we predict only for drivers who start. Only 12 fall in validation and test, so this mainly affects training (podium rate 12.5% → 13.5%). The 381 Indianapolis 500 rows were removed as well, because that race's field was almost entirely separate from the championship.
- **Imbalance.** About 6.4 non-podium rows per podium row in training.

7. Preprocessing

Three steps, all **learned from training rows only** and then applied to validation and test inside a scikit-learn pipeline: a cap on `quali_gap_pct` at the training 99th percentile (11.54%), median imputation, and standard scaling. Gradient boosting uses only the cap; it handles missing values natively and trees ignore scale.

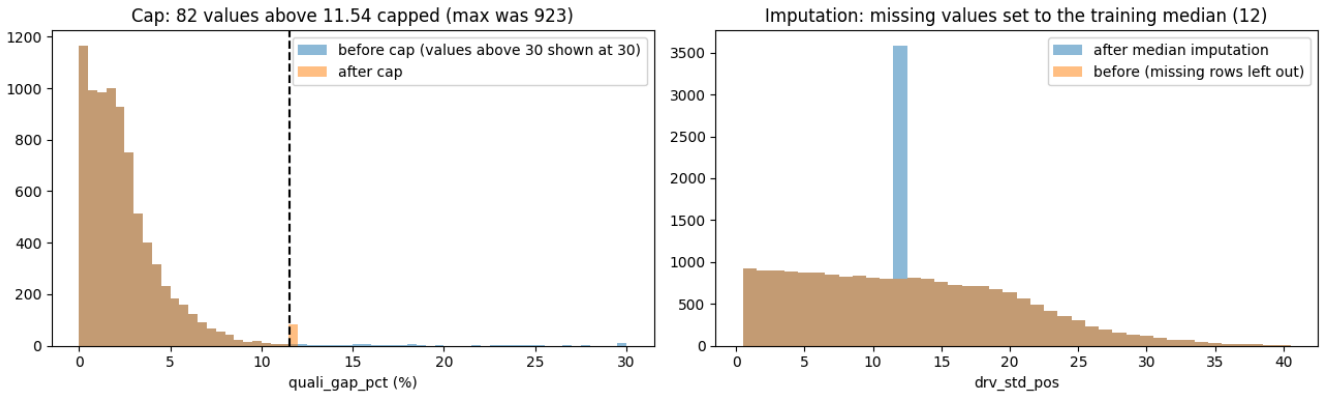


Figure 13. Left: the cap moves 82 extreme values (maximum 923%) to 11.54% and leaves the rest unchanged. Right: median imputation sets about 2,780 missing championship positions (12.4% of training rows, mostly round 1) to 12, creating a spike.

Scaling. Before scaling, feature spreads ranged from 0.04 (`grid_is_zero`) and 0.21 (`drv_podium_last10`) to 203.6 (`con_races_before`); afterwards every feature has mean 0 and standard deviation 1. This lets logistic regression's regularisation treat features equally and helps the neural network train. The imputation spike is a cost: a reigning champion at round 1 looks like a mid-field driver to the linear models.

Training-window experiment

Because old seasons differ from the test era (Figure 10, Figure 5), we tested whether training only on recent seasons helps. Same features and preprocessing; validation PR-AUC.

First training season	Rows	Baseline (grid only)	Logistic regression	Gradient boosting
1950	22,459	0.648	0.733	0.740
1994	10,093	0.648	0.724	0.732
2003	6,783	0.648	0.728	0.708
2014	2,471	0.648	0.720	0.694

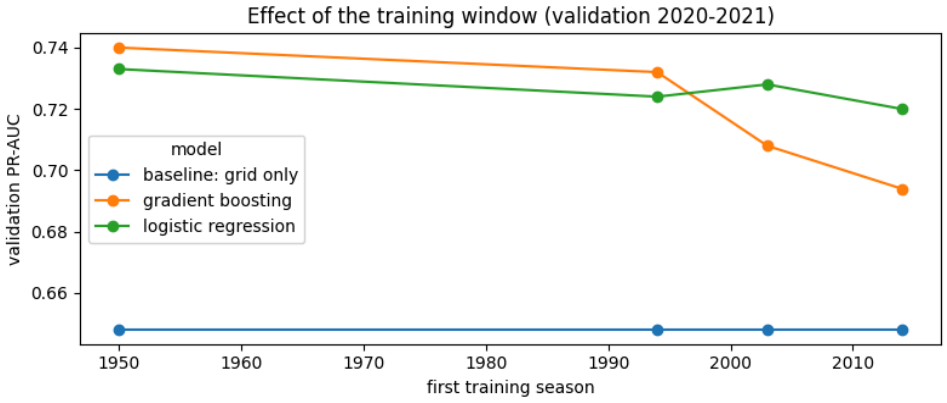


Figure 14. Validation PR-AUC by first training season.

Shorter windows never helped; gradient boosting lost most (0.740 → 0.694) because it needs many rows. The baseline is identical in every window: with one feature its ranking of drivers is always the same, and PR-AUC depends only on that ranking. **Decision:** all models train on 1950–2019.

8. Models and evaluation

8.1 Models

- **Baseline: logistic regression on grid position alone**, the simplest reasonable model. Every other model must beat it.
- **Logistic regression** (17 features): a weighted sum of the features turned into a probability. Simple and interpretable, but only straight-line effects.
- **Gradient boosting** (HistGradientBoosting): a sequence of small decision trees, each correcting the previous ones. Learns interactions and handles missing values natively.
- **Shallow neural network** (scikit-learn MLP): one or two hidden layers of units that combine the features non-linearly.

All models use the same rows and the same 17 features. 22 candidate settings were compared on validation PR-AUC: logistic regression C $\in$ {0.01, 0.1, 1, 10}; gradient boosting learning rate $\in$ {0.05, 0.1} and max leaves $\in$ {15, 31} with early stopping; neural network hidden layers $\in$ {(16), (32), (32, 16)} and alpha $\in$ {0.0001, 0.01}. Logistic regression and boosting were also tried with and without class weights. The F1 threshold was then chosen on validation (0.05–0.95).

- Without early stopping, gradient boosting overfit (best 0.730, below the untuned default at 0.740); with it, it reached 0.742.
- Logistic regression was insensitive to C (0.731–0.733). The neural network needed at least 32 units (16 units: 0.667–0.686).
- **Class weights (balancing experiment)** lowered PR-AUC slightly (gradient boosting 0.742 → 0.731; logistic regression 0.733 → 0.726) and left the best F1 almost unchanged; weighted models simply needed a higher threshold (about 0.75 instead of 0.2–0.4). Threshold tuning does the same job, so class weights were not used.
- **Selected model: gradient boosting** (learning rate 0.05, 31 leaves, early stopping, threshold 0.34), chosen on validation before any test evaluation. The three models differed by at most 0.009 on validation, within the noise of 117 podiums.

8.2 Metrics

PR-AUC is the primary metric: podiums are rare (about 15%), and PR-AUC measures how well true podium finishers are ranked at the top. **ROC-AUC** (overall separation) is secondary. **F1** describes the yes/no decision at the validation threshold. The brief's Section 4 lists ROC-AUC, PR-AUC and F1; we also report **precision** and **recall**, the two parts of F1, to make five.

8.3 Results

Model	Set	ROC-AUC	PR-AUC	F1	Precision	Recall
Gradient boosting (selected)	train	0.938	0.685	0.650	0.589	0.724
Neural network	train	0.906	0.590	0.593	0.512	0.705
Logistic regression	train	0.901	0.575	0.575	0.564	0.586
Baseline (grid only)	train	0.883	0.500	0.544	0.547	0.541
Gradient boosting (selected)	validation	0.935	0.742	0.717	0.708	0.726
Neural network	validation	0.930	0.739	0.697	0.677	0.718
Logistic regression	validation	0.930	0.733	0.703	0.689	0.718
Baseline (grid only)	validation	0.910	0.648	0.695	0.698	0.692
Gradient boosting (selected)	test	0.938	0.701	0.679	0.638	0.725
Logistic regression	test	0.936	0.720	0.688	0.651	0.730
Neural network	test	0.932	0.692	0.655	0.597	0.725
Baseline (grid only)	test	0.913	0.647	0.649	0.650	0.647

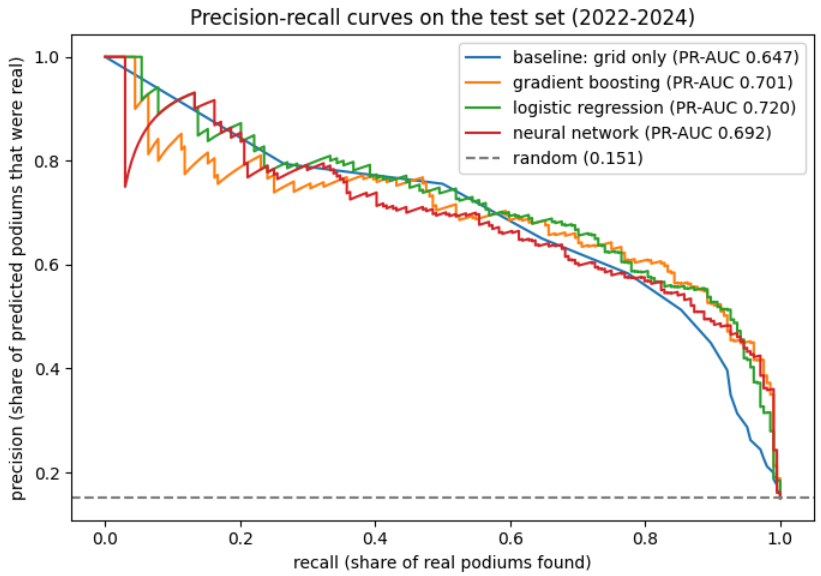


Figure 15. Precision-recall curves on the test set (2022–2024); the dashed line is a random ranking (0.151).

- **All three models beat the baseline on every test metric** (PR-AUC +0.05 to +0.07). The baseline keeps up on the safest picks (front-row starters) but falls away beyond recall 0.7, where the models still find podium finishers who start further back.
- In an average race, gradient boosting flags about 3.4 drivers, of whom about 2.2 finish on the podium; the baseline (in effect the top three on the grid) gets about 1.9.
- **The validation order did not hold on test.** Logistic regression was best on test by 0.019. We kept the model selected on validation; no reliable winner exists among the three, and the simplest model generalised slightly better.
- **Train scores are lower than test scores for every model, including the baseline (0.500 vs 0.647).** This reflects older seasons being harder to predict, not underfitting. Gradient boosting's much higher train score (0.685 vs 0.575–0.590) without a test advantage shows it fits the training seasons more closely.

- **Era sensitivity.** From validation to test the baseline is unchanged (0.648 → 0.647), while the feature-based models drop (logistic regression −0.013, gradient boosting −0.041, neural network −0.047). A plausible cause is the 2022 regulation change, after which form built on 2021 results was temporarily misleading. Recall carried over (about 0.725); precision fell (gradient boosting 0.708 → 0.638).

8.4 Model limitations

- **Logistic regression:** one straight-line effect per feature, so no interactions unless built by hand; needs imputation and scaling.
- **Gradient boosting:** overfits without early stopping; fits the training seasons closely and lost 0.041 PR-AUC from validation to test.
- **Neural network:** sensitive to small input and training changes, no class-weight option, hardest to interpret, largest validation-to-test drop (−0.047).
- **All models:** each driver is predicted independently, so probabilities in a race are not forced to sum to three; weather, safety cars, collisions and failures are invisible before the race; probability calibration was not evaluated.

9. Experiments

9.1 Feature-group ablation

Each group was removed in turn and the model retrained with the selected settings. The validation drop is the result; test is shown for information only and was not used for any decision.

Group removed	GB val drop	LR val drop	GB test drop (info)	LR test drop (info)
Grid (3 features)	0.033	0.008	0.054	0.019
Qualifying (4)	0.007	0.010	−0.024	0.005
Form (6)	0.005	0.004	0.009	0.016
Standings (4)	0.004	0.003	0.035	−0.001

Drop in PR-AUC relative to the full models (GB 0.742 validation / 0.701 test; LR 0.733 / 0.720).

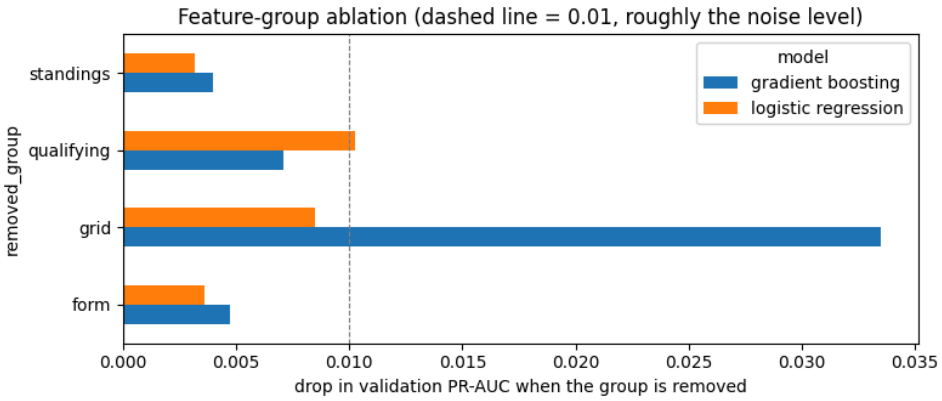


Figure 16. Drop in validation PR-AUC when each group is removed; the dashed line (0.01) is roughly the noise level.

- **Grid is the only group whose removal clearly exceeds the noise.** Without it, gradient boosting falls to 0.646 on test, the level of the grid-only baseline.
- Logistic regression loses little without grid because qualifying position stands in for it (correlation 0.975; qualifying covers about 100% of validation and test rows).
- Form, standings and qualifying each cost at most 0.010 when removed alone, although together they lift PR-AUC from 0.648 to about 0.74. They overlap (form and standings both describe current competitiveness), so one-at-a-time ablation understates their joint value.
- **Decision:** all 17 features kept; no group lowered validation performance. This agrees with the pre-training evidence (grid strongest alone) and adds that the other groups are largely interchangeable.

9.2 Preprocessing-order experiment 1: fit preprocessing before vs after the split

Model	Ours: val PR-AUC	Leaky: val PR-AUC	Ours: test (info)	Leaky: test (info)
Logistic regression	0.7326	0.7328	0.7200	0.7200
Neural network	0.7392	0.7326	0.6920	0.6844

Learning the medians, scaling and cap from all rows (train + validation + test) moved them towards the modern era (cap 11.54 → 12.45; median `quali_gap_pct` 1.961 → 1.836; median `drv_races_before` 43 → 48), so information from the evaluation seasons did enter. The effect on scores is negligible because training is 91% of the rows; the neural

network's change is training noise rather than a leakage gain. We keep the correct order on principle; with a smaller training set the leak would be larger.

9.3 Preprocessing-order experiment 2: build features before vs after removing non-starters

Model	Ours: val PR-AUC	Other order: val PR-AUC	Ours: test (info)	Other order: test (info)
Gradient boosting	0.7421	0.7383	0.7008	0.6866
Logistic regression	0.7326	0.7329	0.7200	0.7208

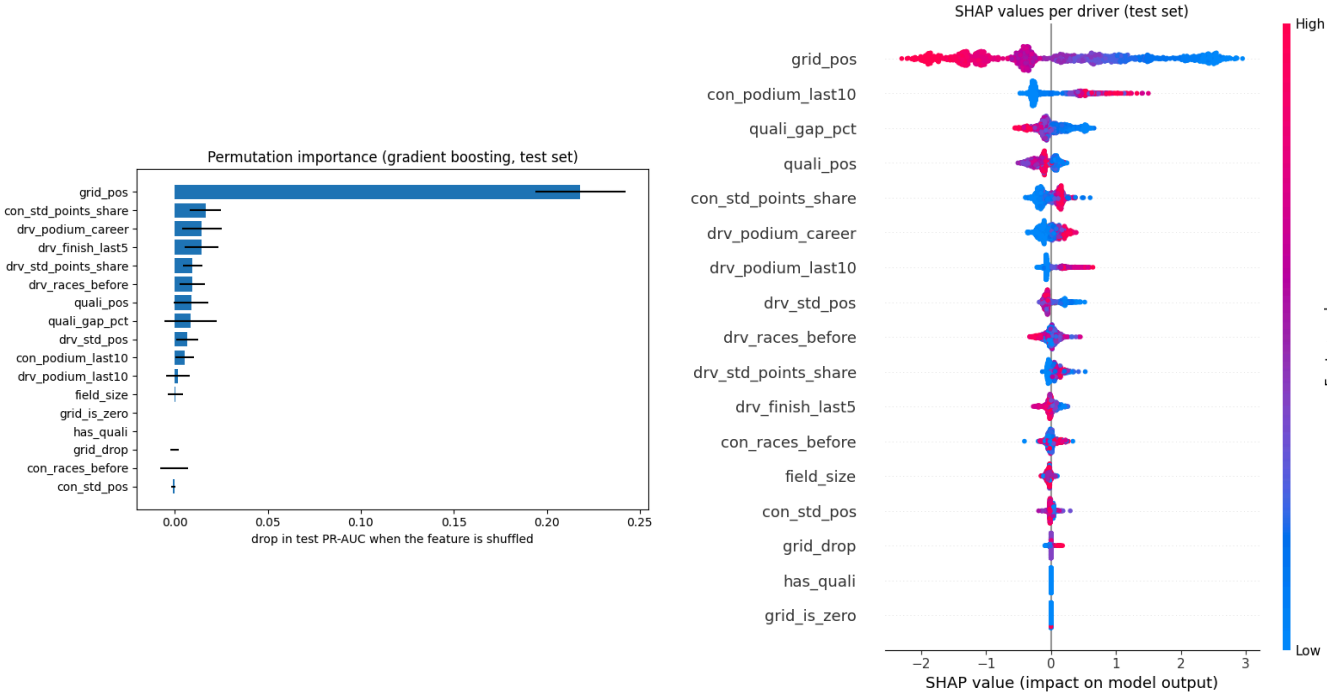
Removing non-starters first changes many feature values (race counts 54–59% of rows, career podium rate 39.6%, team form 15.9%, last-5 finishing order 10.9%, driver last-10 podium rate 1.9%) but has little effect on scores: the changes sit in older seasons, where failing to qualify was common, and in the weakest features. Gradient boosting is slightly better with our order on both validation and test. We keep it, because a failed qualifying attempt is genuine pre-race history.

10. Explainability

> **NOT CAUSAL**
> Importance shows what the model **relies on** to predict, not what **causes** a podium. Grid position is the clearest pre-race trace of car speed: a fast car qualifies well and finishes well. Moving a slow car to pole would not make it a podium contender.

10.1 Global explanations (gradient boosting, test set)

Permutation importance shuffles one feature and measures the drop in PR-AUC (20 repeats). **SHAP** splits every prediction into per-feature contributions in log-odds; the mean absolute value gives a global ranking. The test set is used because it is the larger held-out set (204 podiums); nothing here changes the model.



Figures 17–18. Left: permutation importance with standard deviations. Right: SHAP beeswarm, one dot per driver; colour is the feature value (red high, blue low), position its effect on the prediction.

Feature	Permutation (PR-AUC drop)	Rank	Mean |SHAP|	Rank
`grid_pos`	0.218	1	1.134	1
`con_podium_last10`	0.005	10	0.352	2
`quali_gap_pct`	0.008	8	0.176	3
`quali_pos`	0.009	7	0.153	4
`con_std_points_share`	0.017	2	0.141	5
`drv_podium_career`	0.015	3	0.120	6
`drv_finish_last5`	0.014	4	0.052	11
`has_quali`, `grid_is_zero`	0.000	13	0.000	16

- **Both methods rank grid position first by a wide margin**; shuffling it lowers test PR-AUC from 0.701 to about 0.48.
- **Below grid the rankings differ** (Spearman 0.68) because of redundancy. Team form moves predictions strongly (SHAP 2nd), but shuffling it barely hurts (permutation 10th) because the team's championship points share carries the same information. Permutation importance measures what the model cannot do without; SHAP measures what it actually uses.
- `has_quali` and `grid_is_zero` have zero importance: gradient boosting handles missing values itself, and grid 0 is already encoded in `grid_pos`.
- **Directions are sensible:** starting near the front, strong recent team and driver results, a high championship position and a small gap to pole raise the prediction. A positive `grid_drop` (starting behind one's qualifying position) also raises it slightly, consistent with Question 1 (28.9% podium rate for penalised front-row qualifiers).

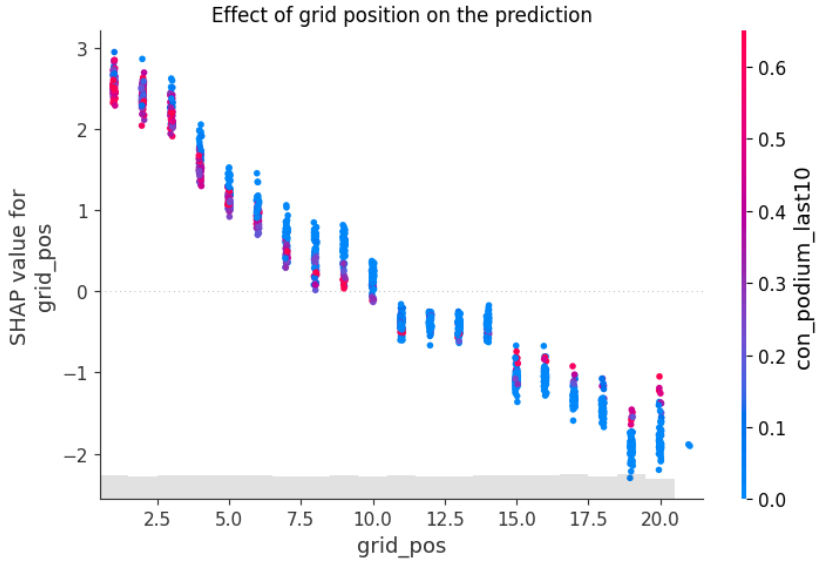


Figure 19. SHAP contribution of grid position, coloured by team form. The effect falls from about +2.5 at pole to about 0 at 10th and −2 at 19th–20th. At the back of the grid strong teams (red) are penalised less (about −1.2 vs −2.0), reflecting fast cars recovering from penalties or pit-lane starts.

10.2 Local explanations (SHAP waterfall and LIME)

Three test-set predictions: the most confident correct podium, the podium finisher rated lowest, and the non-podium driver rated highest. SHAP starts from the model's average output (−3.29 log-odds, 3.6%) and adds each feature's contribution. LIME fits a simple model around the driver; its R^2 shows how well it imitates the real model there.

Case	Driver, race	Predicted	Finished	Main SHAP contributions	LIME R^2
Correct podium	Verstappen, 2023 Dutch GP	85.9%	P1	grid 1 +2.58, team form 0.75 +0.55, gap to pole 0% +0.54	0.52
Missed podium	Gasly, 2024 São Paulo GP	0.5%	P3	gap 7.2% −0.46, grid 13 −0.36, team form 0 −0.27	0.11
False alarm	Verstappen, 2022 British GP	83.7%	P7	grid 2 +2.49, team form 0.65 +0.52, gap to pole 0% +0.52	0.53

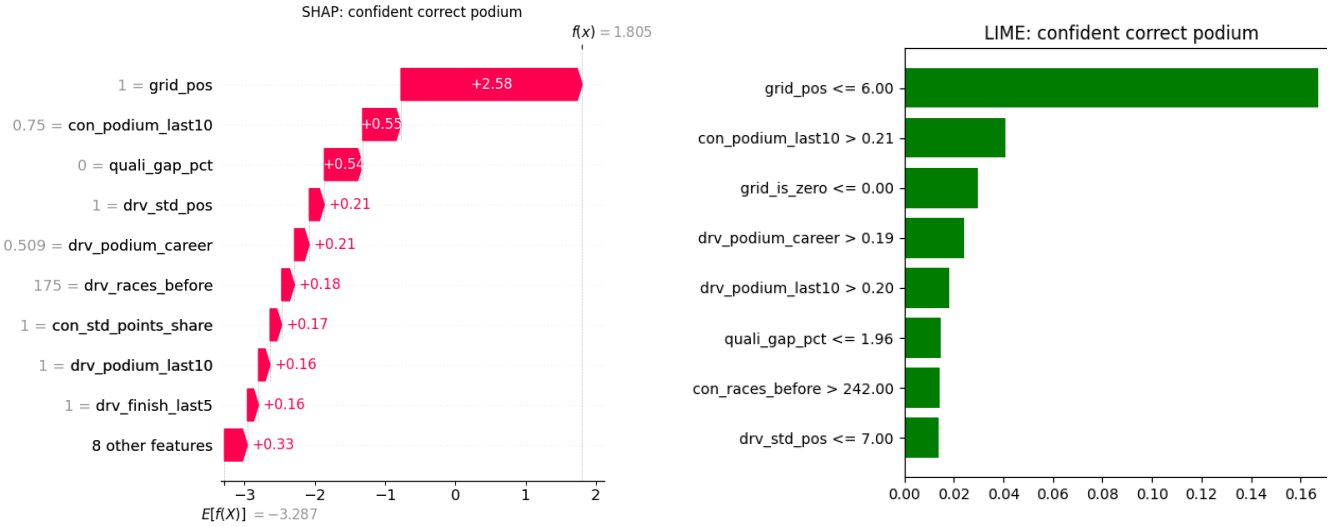


Figure 20. Correct podium: SHAP waterfall (left) and LIME (right). Every feature pushes towards a podium.

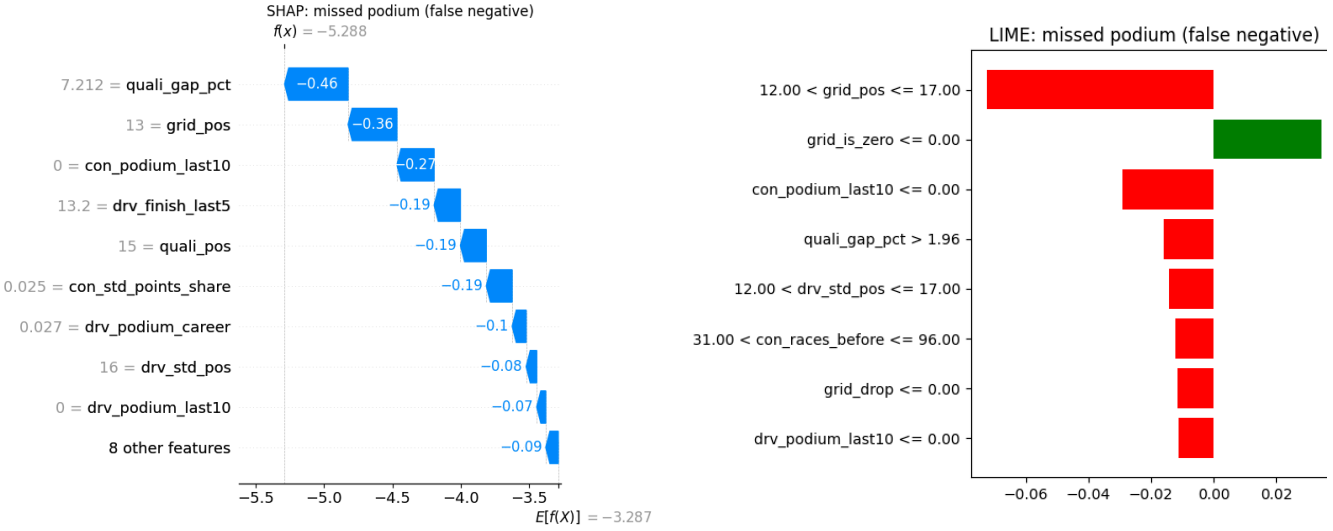


Figure 21. Missed podium: every feature pushes away from a podium. LIME's local fit is weak here (R^2 0.11).

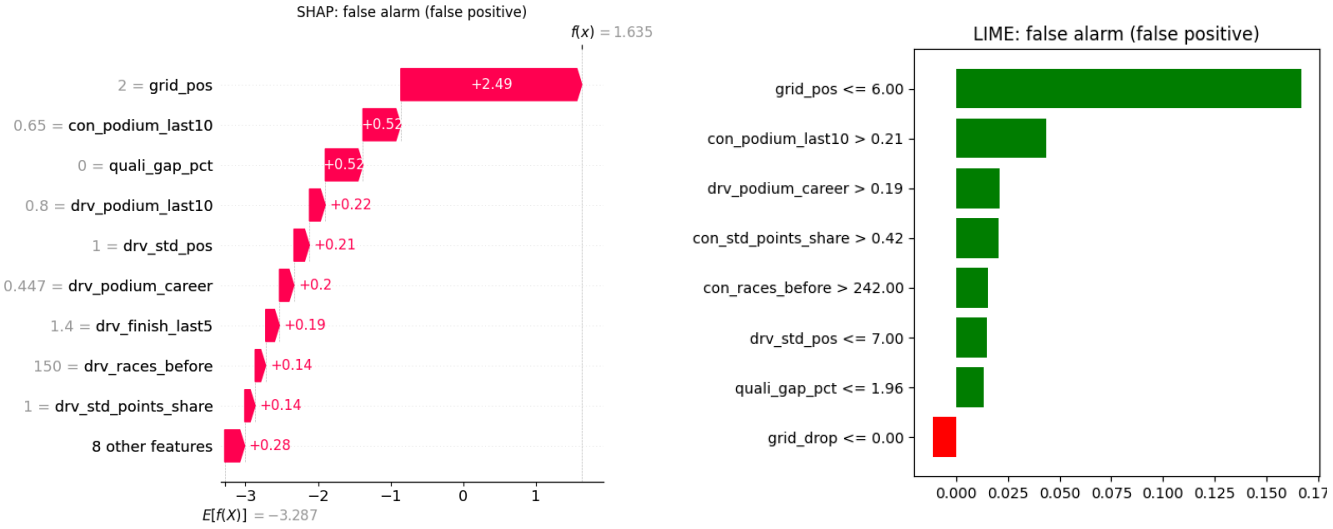


Figure 22. False alarm: almost identical to Figure 20.

- **The reasoning is sensible in every case:** grid position dominates, followed by team form and qualifying pace. SHAP and LIME agree on the order of the main features (their units differ: log-odds vs probability).

- **Both errors come from race-day events.** The 2024 São Paulo GP was a wet race in which Verstappen won from 17th and both Alpines reached the podium; the pre-race data contains nothing about weather or incidents. At the 2022 British GP, Verstappen's pre-race profile was almost identical to his 2023 Dutch GP win, but debris damaged his car's floor and he finished 7th. A prediction around 84% is expected to fail about one time in six.
- **LIME is unreliable where the model's output is flat** (prediction near 0%, R^2 0.11), so for that case we rely on the exact SHAP explanation.

11. Inference function

`predict_podium(record)` takes one driver's raw pre-race inputs and returns the podium probability, the yes/no prediction at the validation threshold (0.34), the list of missing inputs, and the three features that moved the prediction most (local SHAP).

- The caller supplies raw values (e.g. `grid = 0`); the function builds the 17 features with the training recipe (grid 0 behind the last car, gap capped at 11.54, `grid_drop` and `has_quali` derived).
- `grid` and `field_size` are required: they were never missing in training, so the model never learned to handle their absence, and both are always known after qualifying. All other inputs are optional (3–66% missing in training).
- Post-race inputs (points, finishing position, status, …) and unknown names are rejected with an error.

Record	Probability	Prediction	Missing inputs
Complete (Verstappen, 2023 Dutch GP)	85.9%	podium (identical to the evaluation)	0
Same driver, qualifying and standings removed	74.2%	podium	6
Debutant: grid 9, team information only, `drv_races_before = 0`	7.6%	no podium	9
Complete record plus `points`	–	refused (leakage)	–

Missing values are not neutral. In training, missing qualifying data mainly meant a pre-1994 race, and missing constructor standings meant round 1 or a race before 1958. The model reads these gaps as signs of an older race, in which grid position was less decisive. Removing qualifying from a modern pole-sitter lowers his prediction (85.9% → 74.2%), while for the mid-field debutant each missing input adds about +0.10 log-odds. For modern races, qualifying and standings are always available before the start and should be supplied.

12. Known limitations

- **Historical qualifying-coverage drift.** No qualifying data before 1994, complete only from 2003: 36.9% of training rows have it against about 100% of validation and test rows. The model treats missing qualifying as a sign of an older race (Section 11).
- **Era sensitivity.** Grid position became more decisive over time (pole 53.0% → 80.7%); the feature-based models lost 0.013–0.047 PR-AUC from validation to test, plausibly because of the 2022 regulation change.
- **The pre-race cutoff sets a ceiling.** Weather, safety cars, collisions and mechanical failures happen after the cutoff (both local-explanation errors).
- **Small evaluation sets.** Validation has 117 podiums and test 204, so differences of about 0.01–0.02 between models are not reliable; the validation winner was not the test winner.

- **Duplicate rule.** For at most 85 driver-race pairs (0.3%, all before 1979) the kept row's grid and constructor may belong to the car the driver took over; 18 races have 4–5 podium rows.
- **Team identity.** Constructors follow `constructorId`, so a renamed team starts its history from zero (in form features and in Question 3).
- **Missing standings.** No constructor standings before 1958 and none at round 1; median imputation turns these into a spike for the linear models.
- **Data-engineering caveats.** Small per-circuit samples (Q1), nationality as a proxy for home (Q2), the unspecific "Retired" status (Q3).
- **Not used or not evaluated.** Sprint results (only from 2021); probability calibration.